

*MA.6.NSO.3.5***Benchmark****MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals and percentages.**

Example: The number $1\frac{5}{8}$ can be written equivalently as 1.625 or 162.5%.

Benchmark Clarifications:

Clarification 1: Rational numbers include decimal equivalence up to the thousandth place.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.1.1
- MA.6.AR.3.4, MA.6.AR.3.5
- MA.7.AR.3.1

Terms from the K-12 Glossary

- Dividend
- Divisor
- Rational Number

Vertical Alignment**Previous Benchmark**

- MA.4.FR.1.2
- MA.5.NSO.1.2
- MA.5.NSO.2.2
- MA.5.FR.1.1

Next Benchmarks

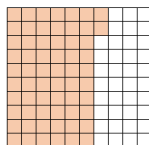
- MA.7.NSO.1.2

Purpose and Instructional Strategies

In previous courses, students related the relationship between decimal values and fractions out of ten, one hundred or one thousand. Students also were taught that fractions show a division relationship. This understanding is being extended in grade 6 accelerated to re-write positive and negative rational numbers into equivalent forms, including terminating and repeating decimals. In future courses, students will generate equivalent forms of rational numbers.

- Instruction includes various methods and strategies to rewrite numbers into a percentage.
 - Finding equivalent fractions with denominators of 10, 100 or 1000 to determine the equivalent percentage.
 - Writing fractions and decimals as hundredths; the term percent can be substituted for the word hundredth.
 - Multiplying the decimal by 100.
- Percent means “per 100,” so a whole is out of 100, or a whole is 100%. If converting the inverse relationship from a percent to a decimal, students divide the percent by 100 to find the equivalent decimal. Students should be encouraged to look for and discover the pattern that occurs (*MTR.5.1*).

- Instruction includes various methods and strategies to rewrite numbers from fraction to decimal or from decimal to fraction (*MTR.3.1*).
 - Decimal grid models



$$0.62 = \frac{62}{100} = 62\%$$

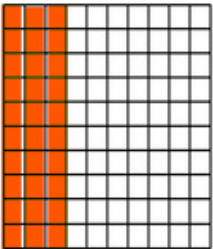
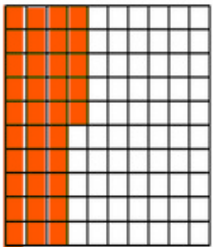
- Dividing the numerator by the denominator.
If the fraction is a fraction greater than one or a mixed number, there will be a whole number in the decimal equivalent and the percent will be greater than 100. Students can convert mixed numbers to fractions greater than one and then divide to help them see the pattern of where the whole number falls in relation to the decimal (*MTR.5.1*).
- Instruction focuses on relating fractions, decimal and percent equivalence to familiar real-world situations, like scores and grades on tests or coupons and sales offered by stores (*MTR.5.1, MTR.7.1*).
- Use questioning to help students determine their solution's reasonableness (*MTR.6.1*).
 - For example, is it reasonable for the percent to be more than 100? Is it reasonable for the percent to be less than 1?
- Instruction includes the understanding that percents are a number and are worth specific amounts in different contexts (*MTR.7.1*).
- Students should have practice with and without the use of technology to rewrite positive rational numbers in different but equivalent forms.

Common Misconceptions or Errors

- Students may incorrectly think that, when dividing, the larger value is divided by the smaller value. This is a common overgeneralization because most of their experience in elementary school is dividing larger values by smaller values.
- Students may misplace the decimal when converting from fraction to decimal form because they forget to place the decimal after the whole number in the dividend if the numbers do not divide evenly or do not place the decimal in the aligned place value of the quotient. It is helpful for some students to use graph paper when dividing with decimals to help students keep place values accurately aligned.
- Students may incorrectly think that the decimal value and the percent are exactly the same, not realizing that the percent is 100 times the decimal value.
- Students may incorrectly convert from fraction to decimal form because of their lack of place value knowledge. Instruction includes students using the expanded form of the numbers by adding a decimal and zeros in aligned place values until the decimal terminates. Practice includes multiple opportunities for students to work with fractions that require students to add zeros.
- Students may incorrectly move the decimal direction when converting between decimals and percentages because they do not understand what happens to the decimal when a number is multiplied or divided by a power of 10.

Strategies to Support Tiered Instruction

- Instruction includes the use of estimation when converting fractions to decimal form to ensure the proper placement of the decimal point in the final quotient.
- Instruction includes the use of a 100 frame to review place value for tenths, hundredths, and if needed, thousandths and the connections for decimal and fractional forms.
 - For example:

Fraction	Model	Decimal
$\frac{3}{10}$		0.3
$\frac{35}{100}$		0.35

- Instruction includes providing multiple opportunities for students to work with fractions that require students to add zeros.

Instructional Tasks

Instructional Task 1 (MTR.7.1)

Kami told her mother that she answered 17 out of 25 questions correctly on her math test or 68%. Did Kami determine the correct percent score? Explain your reasoning.

Instructional Task 2 (MTR.2.1, MTR.4.1, MTR.5.1)

Complete the table to identify equivalent forms of each number. Explain how you approached your solutions.

Fraction	Decimal	Percent
		79%
$\frac{3}{8}$		
	1.236	
$2\frac{1}{4}$		
	0.625	

Instructional Task 3 (MTR.2.1)

Convert each of the following to an equivalent form to compare their values.

$\frac{1}{5}$ 0.4 65% $2\frac{3}{4}$ 5.75 $\frac{9}{8}$ 123%

Instructional Items

Instructional Item 1

How can $\frac{3}{4}$ be written as a decimal and as a percent?

Instructional Item 2

What is an equivalent fraction and decimal representation of 42%?

Instructional Item 3

Complete the following statement:

$4\frac{2}{5}$ can be written as the decimal _____ and is equivalent to _____ percent.

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*